

# MagicCube: A One-Handed Interaction Approach in 3D Environment on Smartphones

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## ABSTRACT

When users have only one free hand to operate smartphones, interaction with 3D virtual environment (VE) is restricted. We propose MagicCube, a cube with 5 DOF that perfectly integrates navigation and selection in 3D VE, to address this problem. Three sets of opposing faces are assigned to feature faces with translation, rotation, and selection. Any operation can be performed by dragging the corresponding face. Further, by rotating the cube, users can choose the displayed faces and adjust their relative position according to personal preference. A comparative analysis showed that it could effectively reduce the finger's visual occlusion as well as improve the accuracy and stability of interaction operations.

**Index Terms:** Human-centered computing—Interaction design—Interaction design process and methods—User interface design

## 1 INTRODUCTION

In recent years, the increasing screen size of smartphones and the dramatic development of processor performance have brought obvious optimization to the mobile devices' 3D browsing experience. At present, most of the 3D applications on smartphones use a two-handed interaction design. However, A study found that people often prefer to use just one hand for operating handheld devices in many situations [3]. Therefore, if 3D applications can get rid of the dependence on the two-hands operation, it will be more conducive to its popularity.

Problems such as the unreachable area exist for both landscape and portrait modes when people hold the phone with one hand. The overarching goal of this research is to design, implement, and evaluate a novel interaction approach to accomplish accurate and convenient interaction in 3D VE with one-handed on mobile phones.

## 2 RELATED WORK

In current 3D applications, translation and rotation approaches usually include directional keys, virtual joystick, and sliding gestures. However, both piled UI components and frequent finger movement can lead to the visual occlusion. Meanwhile, The common selection technology in 3D applications is Direct Touch. Although it is easy to apply and has fast response speed, study found that it too error prone. Besides, none of the above solutions work well in one-handed interaction situations.

To the best of our knowledge, in one-handed interaction, there is less attention to 3D applications. But some technical achievements have brought us inspiration. For example, ExtendedThumb technology [4] and a new virtual collaboration environment interaction

of metaphor based on ray-casting [1] jointly support the selection function in MagicCube.

## 3 DESIGN IMPLEMENTATION

Existing studies had shown that the most common interaction requirements in 3D VE include navigation, selection, and manipulation [2]. Among them, manipulation is mostly applied to a specific object after the selection step. Therefore, we determined that essential interactive requirements in 3D VE are translation, rotation, and selection.

Based on the previous analysis of the existing interactive interface, our research goal is to optimize the following four aspects: First, minimize the number of UI components to decrease their proportion of the screen. Second, simplify the complexity of operations. Third, improve the adaptability to one-handed operation. Last, improve the flexibility of the interface.

We planned to find a suitable 3D model as an interactive medium. The cube was chosen because it has at least one face always visible, and three at most. This feature matches precisely the number of functions we need in 3D VE.

First, we placed a cube in the scene and set a camera that only rendered it. Referred to popular mobile games, we temporarily put this camera's viewport on the left. We divided the six faces of a cube into three groups. Each opposite faces belonged to one group, corresponding to translation, rotation, and selection, respectively. Although the operations are different in 3D VE, with specific mapping rules, we could associate them with the displacement vector in the 2D screen space. Therefore, all faces could share one operational method, dragging, same with the use-pattern of virtual joystick. We also added a variable-style ring UI around the cube to indicate a limited drag range. The detailed designs of the functions are as follows:

**Translation:** A user can touch the translation face and drag the cube in any direction to translate in 3D VE. As the finger holds down, the color of the ring changes, and a "Move" prompt appears above as seen in Fig. 1 (left). Displacement vectors of the finger are used to drag the MagicCube and direct the virtual camera's translation. Mapping rules are as follows: the X-axis and Y-axis of the screen coordinate system map to the X-axis and Z-axis of the virtual camera's coordinate system in 3D VE.

**Rotation:** The principle of rotation is much similar to the translation. Displacement vectors are used to calculate the virtual cameras rotation around its own Y-axis and X-axis in 3D VE. The interface changes as shown in Fig. 1 (middle).

**Selection:** The proportion between the whole screen and MagicCube's movable area is used as an amplification factor to multiply displacement vectors, like a virtual thumb. We draw a ray to describe the extended thumb and put a little ring to show the hit point. With ray-casting technology, we can generate the corresponding ray in the 3D VE according to the ring's projection position. Once the ray passes through an alternative object, we set that as a candidate item and change the mapping rule to mitigate ray jitter. The interface is shown in Fig. 1 (right).

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Figure 1: Example scenes of three functions in MagicCube Scheme. Including translation (left), rotation (middle) and selection (right).

Last, since the cube has 3DOF in rotation, by swiping anywhere on the screen except the cube, users could decide the number of visible faces and adjust their relative position. This is a feature that can be turned on and off.

## 4 USER STUDY

To test the usability of MagicCube and make a preliminary comparison with current methods, we took the Virtual Joystick method as a comparison. It applies a virtual joystick for translation, as shown in Fig. 2 (left), screen sliding for rotation, and Direct Touch for selection. Eighteen participants attended the user study and the test order for different methods were randomly determined.

The experimental scene was a virtual exhibition hall, in which we placed 15 exhibits with the numbers 1 to 15, with route tips on the ground. Participants were asked to visit the exhibition according to the prescribed route and trigger all exhibits in digital ascending order. Their preferred cube postures in diverse conditions were recorded, too.

For 14 landscape samples and 11 portrait samples, we collected the following data: the task completion time (TCT), the number of operations times(NOO), the number of hand posture changes (NHPC), and the number of error touch(NET), which means when they inadvertently click exhibits during translating or rotating.

## 5 RESULT AND DISCUSSION

As we expected, Virtual Joystick performed better than MagicCube in TCT due to the speed advantage of Direct Touch. We think the fact that users are more familiar with the former had also affected the result. In other respects, the mean NOO was significantly lower for MagicCube than Virtual Joystick in both screen modes. In terms of accuracy, for Virtual Joystick, participants' interaction contained extensive range of hand movements, such as sliding the screen to rotate and clicking exhibits, which led to obvious NET. With MagicCube's advantage of rotation mechanism and ray selection, 16 of 18 participants completed the task with zero error touch, and 2 participants had less than twice. Wilcoxon analysis showed MagicCube significantly performed better than Virtual Joystick.

To verify MagicCube's support for personalization, we analyzed the cube postures with two or three visible faces chosen by participants. Through vertical comparisons of each participant, we found that most of them have stable preference for interface layout. Further, horizontal comparisons between participants demonstrated the

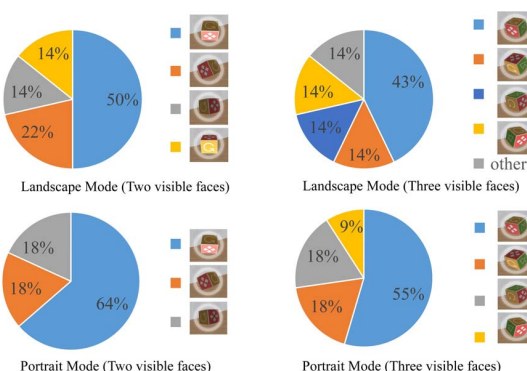


Figure 3: A visualization of MagicCube’s personalized selection.

diversity of preference. Ignoring slight differences, the classification results in both screen modes is illustrated in Fig. 3. This finding is an initial validation of the need for a personalized UI and the potential of MagicCube. In SUS analysis, MagicCube scored high scores in ease of study, support for different functions, and popularity. Some participants felt that some guidance is necessary for the first usage.

## 6 CONCLUSSION AND FUTURE WORK

MagicCube integrates three essential interactive functions in 3D VE into a cube model and supplies them the same operational method. A preliminary study proved it could provide a higher level of visual integrity, interactive stability, operational accuracy, and more UI layout possibilities. In the future, we plan to apply more sophisticated analyses of existing data and design additional experiments to uncover its potential abilities.

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Figure 2: UI of two interactive methods in comparison experiment.